The Learning Wave

Adaptive Thresholding for Bright Spot Detection in Noisy Images

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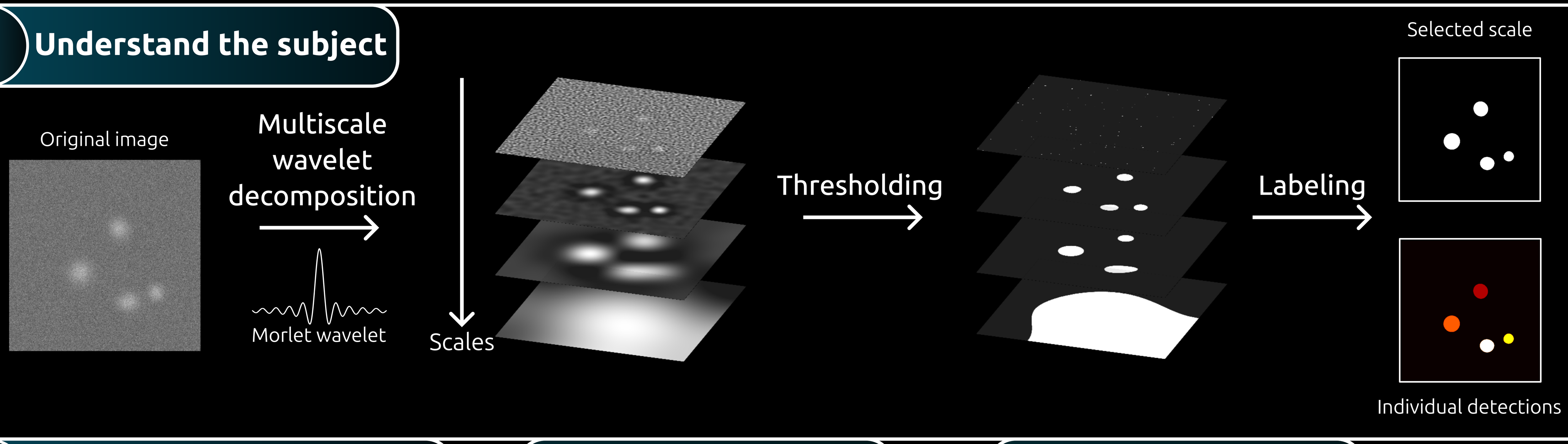
Accurate detection of fluorescent spots in microscopy images is essential for quantitative biological analysis. Wavelet-based approaches, such as the undecimated transform of Olivo-Marin (2002), are widely used due to their ability to capture features across multiple scales and their robustness to noise. However, these methods often rely on global thresholding, which can be suboptimal in heterogeneous images with locally varying noise.

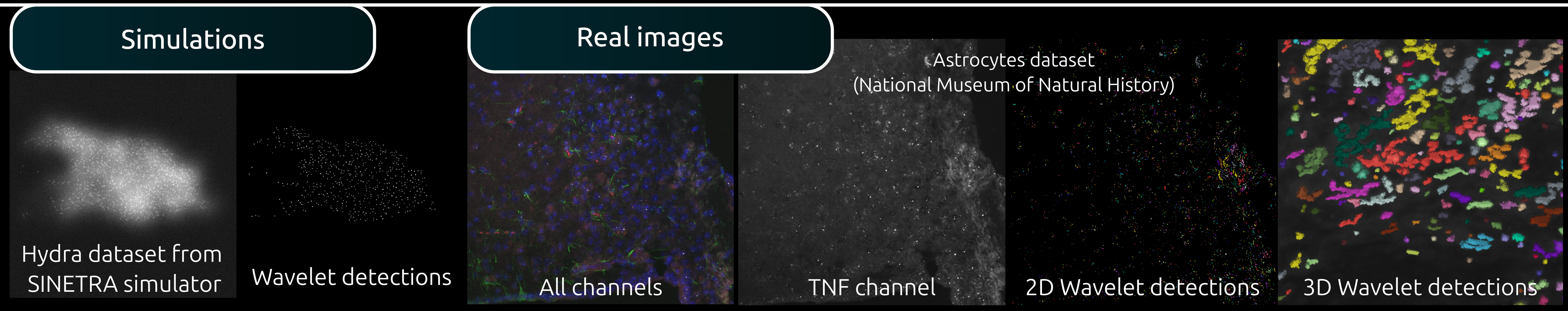
1 Understand the subject



Wavelet-based methods have been widely used for spot detection in noisy images, especially in biological microscopy. The classical approach relies on decomposing the image into multiple scales, applying a threshold, and selecting the most relevant scale. Detected regions are then labeled as individual spots. While effective, this method strongly depends on the choice of the global threshold.

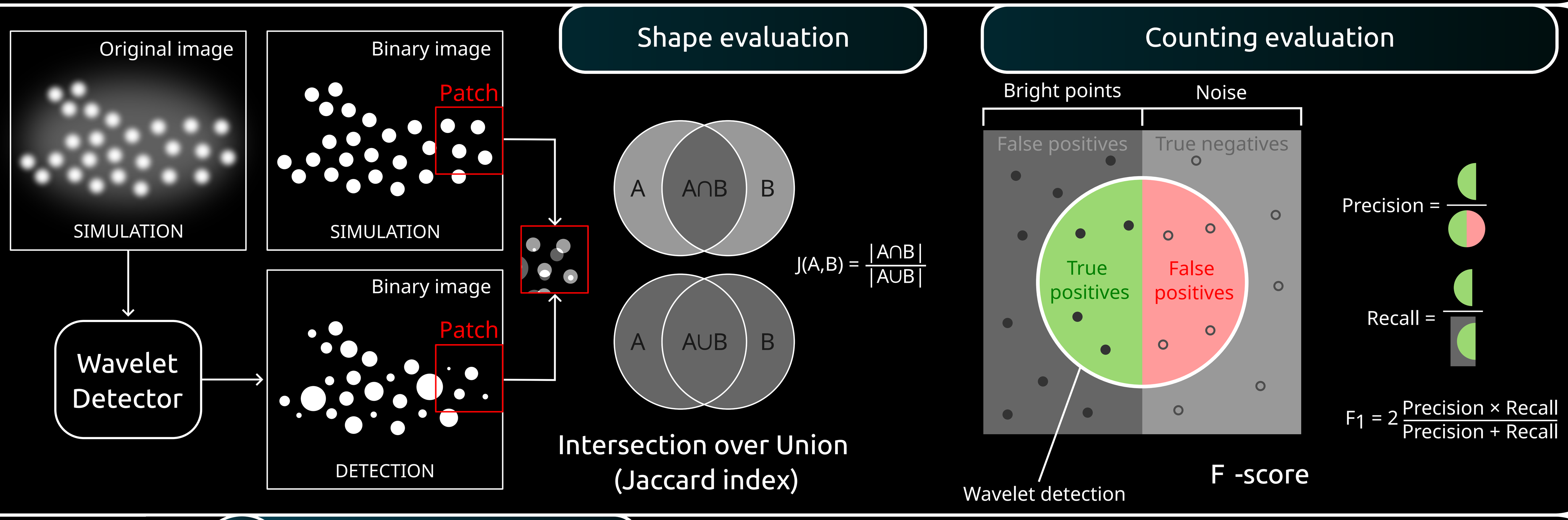
2 Simulated and real images

The advantage of the simulator is that it provides information on the ground truth (the exact position of the points) since everything has been generated artificially. Biological imaging is a useful application of the methods created.

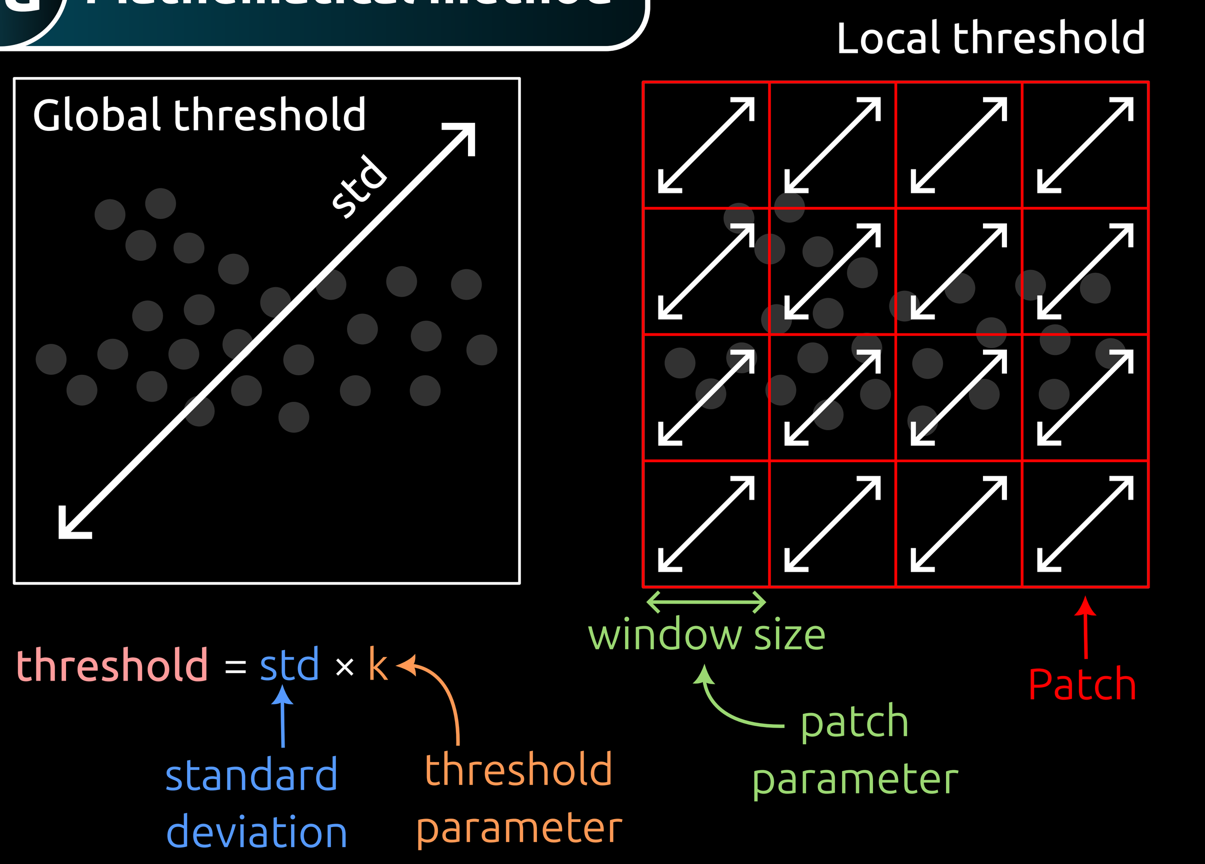


3 Evaluate the detections

Detected spots were compared to ground truth using Intersection over Union (IoU) and the F1-score. IoU quantifies the overlap between predicted and true regions, while the F -score balances precision and recall based on the position of the centroids. These metrics provide a robust evaluation of detection accuracy.

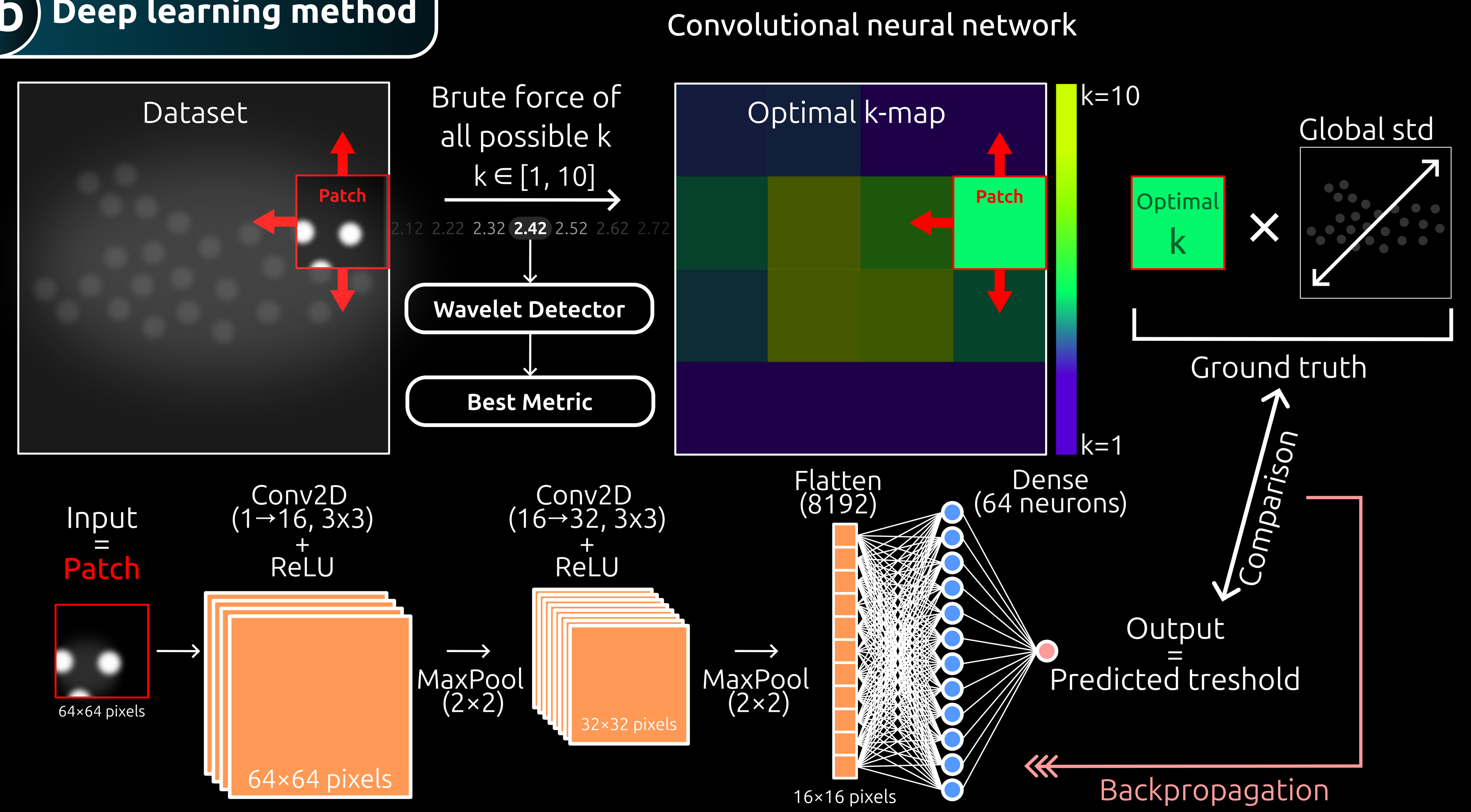


4a Mathematical method

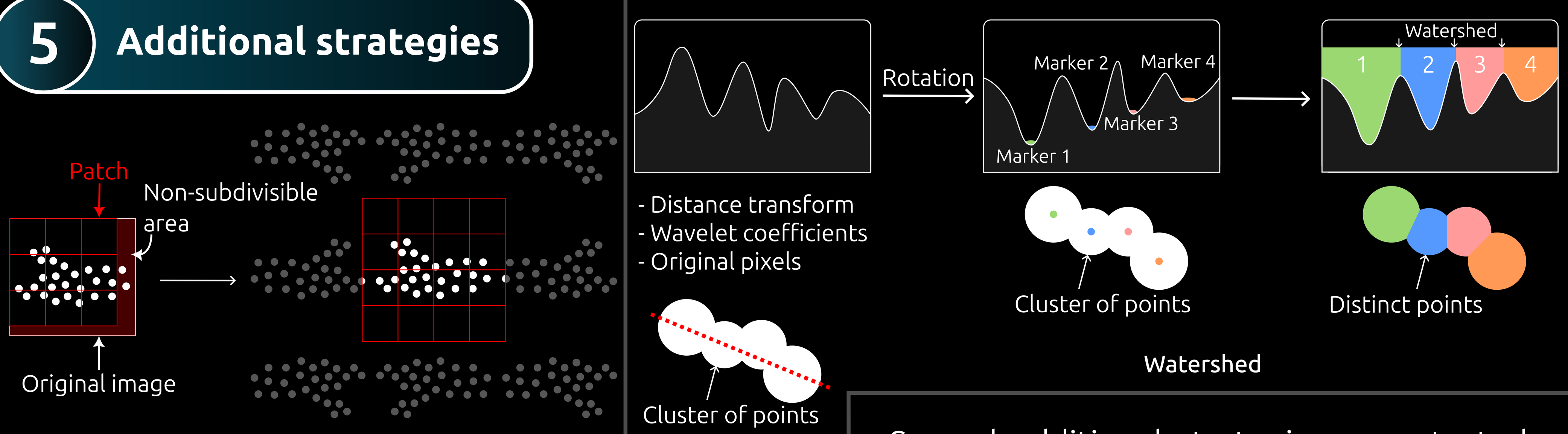


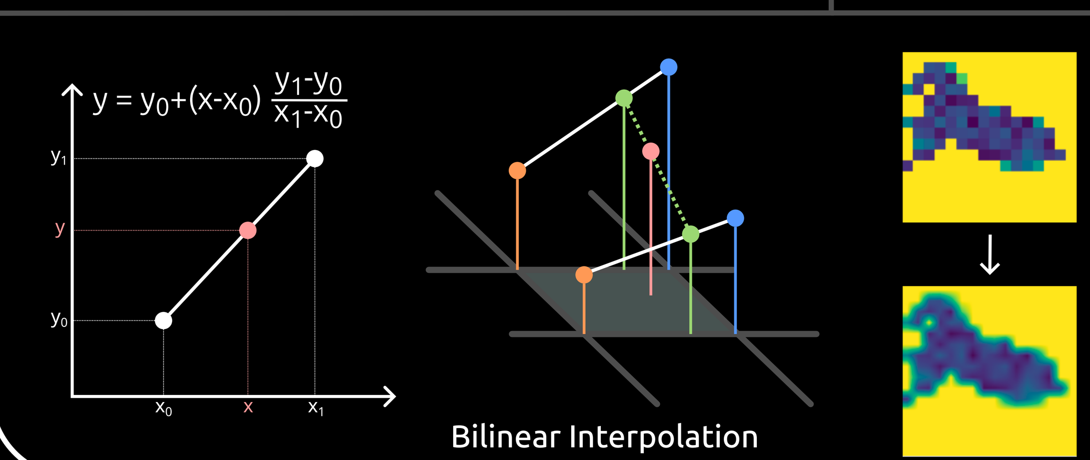
The global threshold rule can fail in heterogeneous images with variable background noise. To address this limitation, a local threshold was introduced: the image is divided into patches, and each patch is assigned its own threshold based on its local standard deviation.

4b Deep learning method



5 Additional strategies





Several additional strategies were tested to improve detection. Interpolation was applied to increase resolution and refine spot localization. Mirror padding was used to reduce border artifacts. Watershed segmentation was employed as a post-processing step to separate overlapping detections and improve instance segmentation.

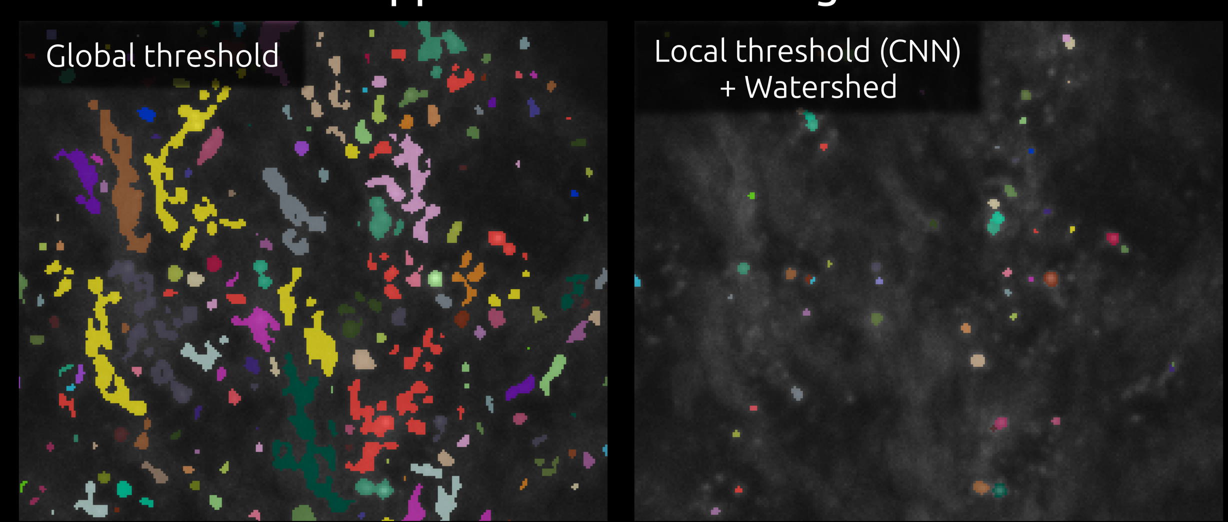
6 Main results

Summary of metrics for the simulator

	Optimal results	Global threshold	Local threshold	Neural network
Jaccard Index	0.663	0.651	0.631	0.628
F_1-score	0.940	0.897	0.905	0.885

Interpolation has no noticeable effect

Application to real images



In real images, the results are much better with an appropriate local threshold (CNN), and the application of watershed to instantiate and count nuclei.

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